UC IPM

< Corn

AGRICULTURE

Grasshoppers

Melanoplus spp.

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Description of the Pest

Grasshoppers can be occasional pests of corn. In late summer and fall, grasshopper eggs are laid in grassy foothills, on ditchbanks, along roadsides and fence rows, in pasture areas, and in alfalfa fields. Eggs hatch in spring and young nymphs feed on nearby plants. When wild grasses and other plants become dry, grasshoppers migrate to irrigated croplands.

Damage

Grasshoppers feed on foliage, most often on the edges of fields near pasture areas or roadsides. They seldom cause economically significant injury.

Management

Topical treatments are most effective; treating field borders may be adequate.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least harmful to natural enemies, honey bees, and the environment listed at the top of the table.** When choosing a pesticide, consider information related to water and air quality, resistance management, and the pesticide's properties and application timing. Use PestManage to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary precautions when handling pesticides.

Expand table

Rank	Active ingredient	Example trade
Application Timing Varies (See *UC IPM Pest Manageme*		
A	carbaryl—XLR formulation *	Sevin XLR Plus
B	malathion	Malathion 8-E

Rank	Active ingredient	Example trade

—	No information.
I	Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.
II	Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to
III	No bee precaution, except when required by the pesticide label or regulations.

- ‡ [a] [b] Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
- * [a] Permit required from county agricultural commissioner for purchase or use.
- [1] ↩ Group numbers for insecticides and miticides are assigned by the Insecticide Resistance Action Committee (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B;

insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.

- [2] ↩ Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- [3] ↩ Risk of harm to honey bees. For more information, see ***Bee Precaution Pesticide Ratings***.
- [4] ↩ Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.
- [5] a b Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.
- [6] ↩ Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- [7] ↩ Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- [8] ↩ Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- [9] ↩ Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).

- 10 ↩ Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 11 ↩ Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 12 ↩ Date information was last updated in the *UC IPM Pest Management Guidelines*.

UC PEER REVIEWED

UC IPM Pest Management Guidelines: Corn

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